Amazon Web Services: Auto-Scaling

Topics

The 3 W's of Scaling

Launch Templates and Launch Configurations

Scaling EC2 Instances with Auto Scaling

The 3 W's of Scaling

What	Where	When
• What do we scale? • We must decide what resources we are going to scale. • How do we define the template?	• Where do we scale? • When applying the model, where does it go? • Should we scale out databases? Or Webservers?	• When do we scale? • How do we know we need more? • CloudWatch alarms can tell us when it is time to add more resources

What is a Launch Template?

- Specifies all the needed settings that go into building out an EC2 instance
- Is a collection of settings you can configure so you don’t have to walk through the EC2 wizard over and over to deploy similar instances

Launch Templates vs. Launch Configurations

Templates

- Newer
- Capable of leveraging all EC2 Auto Scaling features
- Support versioning
- More granularity
- AWS recommends this approach

VS

Configurations

- Predecessor
- Only for certain EC2 Auto Scaling features
- Immutable
- Limited configuration options
- Try not to use them

Demo: Create a Launch Template

Auto Scaling Groups

- An Auto Scaling group contains a collection of EC2 instances that are treated as a collective group for purposes of scaling and management

- What goes into an Auto Scaling group?

1. Define your template
2. Networking and cost
3. ELB Configuration
4. Set scaling policies
5. Notifications

Auto Scaling Restrictions

Minimum	Maximum	Desired
• The lowest number of EC2 instances you'll ever have online • Will not go below this number	• The highest number of EC2 instances you will ever provision • You will not go above this	• How many instances do you want right now?

Demo: Auto Scaling

Checkpoint

What tool can be used to notify you of scaling events?

1. Auto Scaling push messages
2. SQS
3. SNS
4. E-mail

What AWS resource defines the configuration of instances created by EC2 Auto Scaling?

1. Saved configurations
2. Launch settings
3. Instance wizard
4. Launch templates

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